



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

“Cloud Computing- Firebase Cloud Services”

Presented by:

Y. Sravan Kumar(192472289)

CSA1514 :CLOUD COMPUTING AND BIG DATA ANALYTICS

Faculty: Govindasamy C

Date: 20/07/2026



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Introduction

- Firebase is a **Backend-as-a-Service (BaaS)** platform developed by Google.
- It helps developers build web and mobile applications quickly.
- It provides cloud-based backend services without managing servers.
- It is built on **Google Cloud Platform (GCP)**.
- It supports Android, iOS, Web, Flutter, and Unity applications.



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Core Technologies

- **Firebase Authentication** – Secure user login and authentication.
- **Cloud Firestore** – NoSQL cloud database for storing application data.
- **Realtime Database** – Stores and synchronizes data in real time.
- **Cloud Storage** – Stores images, videos, PDFs, and other files.
- **Cloud Functions** – Executes backend code automatically (serverless).
- **Firebase Hosting** – Hosts websites and web applications.
- **Firebase Cloud Messaging (FCM)** – Sends push notifications.



SIMATS
ENGINEERING

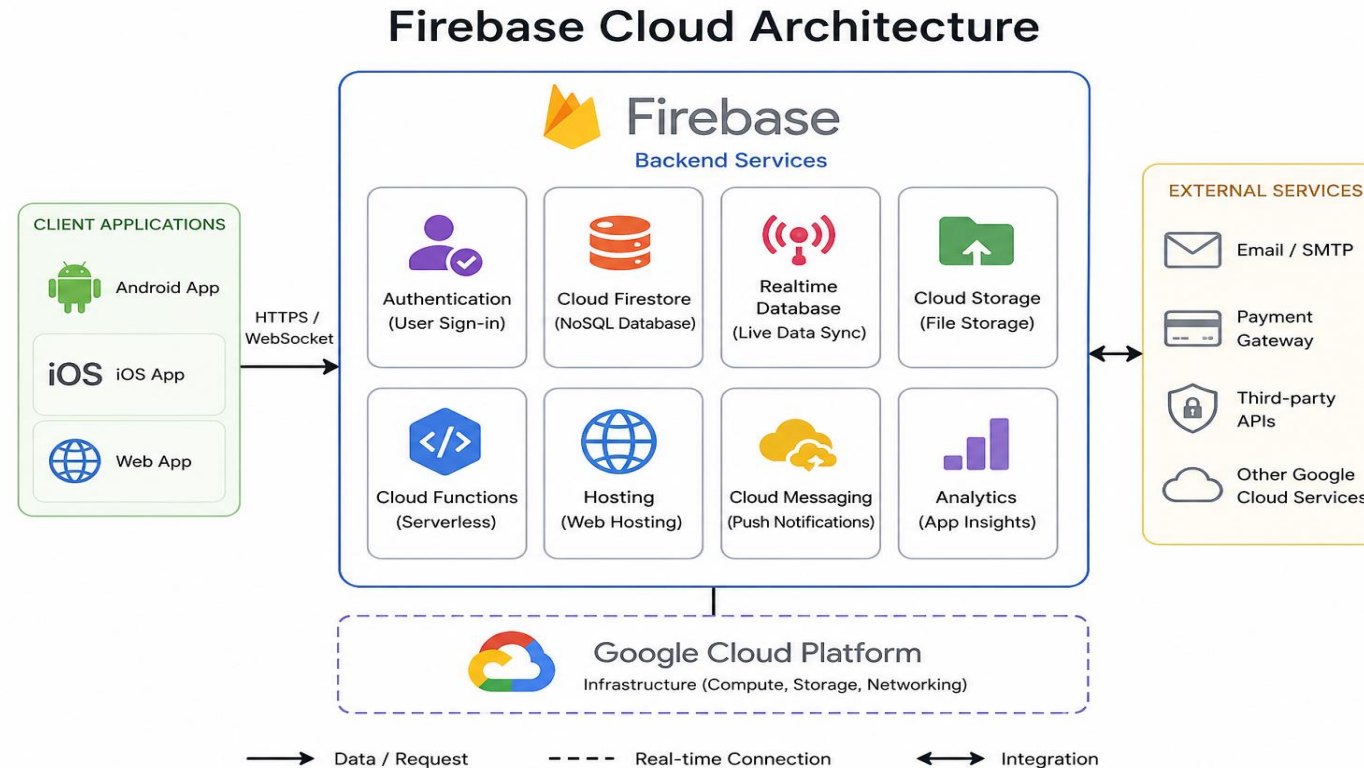


SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Components of Firebase Architecture

- Client Application (Android, iOS, Web)
- Firebase Authentication
- Cloud Firestore / Realtime Database
- Cloud Storage
- Cloud Functions
- Firebase Hosting
- Cloud Messaging (FCM)
- Firebase Analytics
- Google Cloud Platform Infrastructure

Architecture Diagram





SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Benefits

- Fast application development.
- No server management.
- Automatic cloud scaling.
- Secure user authentication.
- Real-time data synchronization.
- Easy integration with Android and iOS.
- Free tier for beginners.
- Reliable and highly available services



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Example Use Case (Online Shopping App)

- User logs in using Firebase Authentication.
- Product details are stored in Cloud Firestore.
- Product images are stored in Cloud Storage.
- Orders are processed using Cloud Functions.
- Customers receive order notifications through FCM.
- Firebase Analytics tracks user activity and sales.



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Conclusion

- Firebase simplifies backend development.
- It provides authentication, databases, storage, hosting, and analytics in one platform.
- It reduces development time and infrastructure costs.
- It is suitable for startups, students, and enterprise applications.
- Firebase continues to evolve with AI, cloud computing, and serverless technologies, making it a powerful platform for modern application development.



SIMATS

ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

THANK YOU



SIMATS

ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

